

TrES-5b

R-0738

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250519 · created 2026-07-28T01:33:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460814.9134 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.023
Transit depth (Rp/Rs)^2	2.94 +/- 0.8 [%]
Orbital Inclination (inc)	83.88 +/- 1.98
Airmass coefficient 1 (a1)	1.1 +/- 0.021
Airmass coefficient 2 (a2)	-0.083 +/- 0.089
Scatter in the residuals of the lightcurve fit is	1.94 %
Best Comparison Star	#1 - [114, 83]
Optimal Method	PSF photometry
Transit Duration (day)	0.066 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.171 ± 0.023 vs 0.142 (deviation 1.26σ)
Duration fit vs published	0.066 d vs 0.0758 d (deviation 0.76σ)
Mid-time O–C	-2.1 min
Depth SNR	7.4
Residual scatter	1.94 %

Light curve

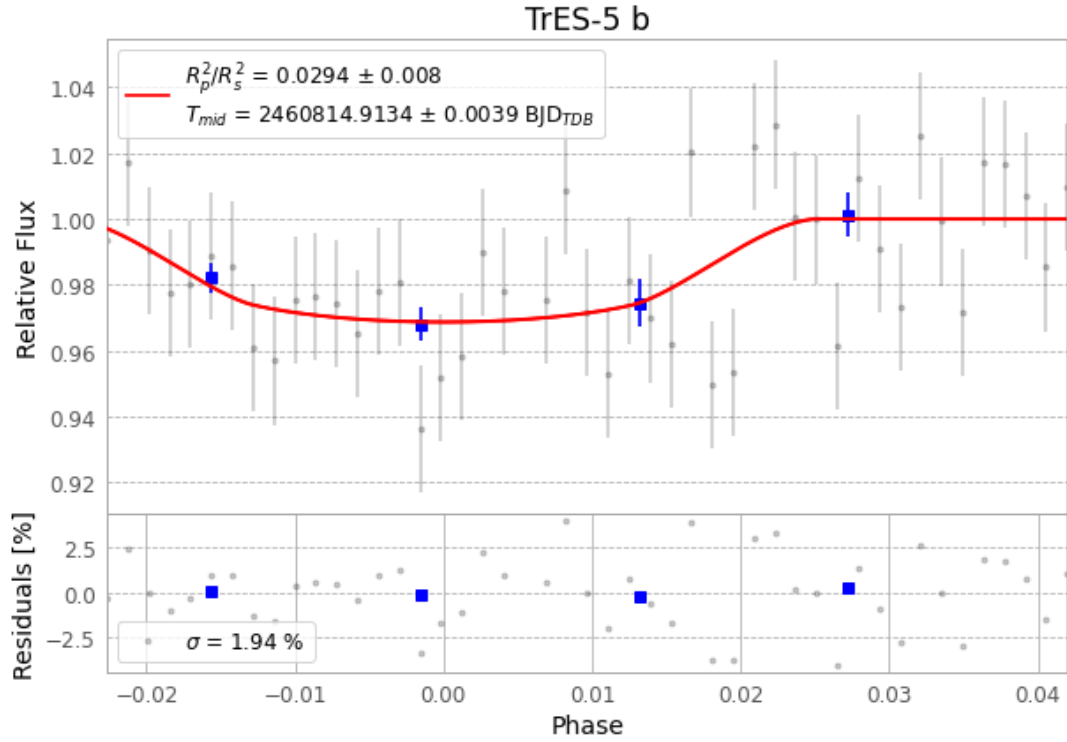


Figure 1 — FinalLightCurve_TrES-5 b_2025-05-19.png (SHA-256 264fcbd581199c5e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 233], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7347e9ddf71f3b8f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

47 FITS frames (31 MB), TRES-5250519090616.FITS → TRES-5250519112416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250519.log (de8644238b6f...), FinalLightCurve_TrES-5 b_2025-05-19.png (264fcbd58119...), FinalLightCurve_TrES-5 b_2025-05-19.csv (ecee024c7b66...)

Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:33Z.